Lesson plan · for the teacher

First programs: print, variables and input

Programming · Level 2 · Lesson 1

Time
40 min

Materials
Board and chalk; Your phone with this lesson downloaded (Python runs offline after one download of 0.29 MB); Paper for pupils' predictions

Learning goals

- Predict what a short program shows, then run it.
- Tell text in quotes from numbers.
- Say what a variable holds after it changes.

Activity: Predict, then run

1. Warm-up (5 min): A pupil plays a robot and walks to the door following only your exact words. What happens with an unclear instruction? Agree: a program is exact instructions, in order.
2. Teach (10 min): Write print("Good morning") and print(7 * 6): text in quotes is shown as written; sums are worked out. Then mark = 14: = stores the value on the right in the name on the left.
3. Predict (10 min): Write print("7 * 6"), print("5" + "3"), and pupils = 30, pupils = pupils + 2, print(pupils). Pairs write their predictions on paper.
4. Run (10 min): Run each program on your phone and read out what it shows. Tally the predictions. A pair who was wrong explains first, then a pair who was right.
5. Wrap-up (5 min): Ask which program surprised the class most, and why. Then give the exit ticket.

One phone is enough: run the programs yourself and read the results aloud, or pass the phone to the pair who predicted best. Download the lesson the day before, so Python works without a connection.

Discussion

- Why does print("5" + "3") show 53?
- How can pupils = pupils + 2 make sense?
- What job could a short program do for our class?

Exit ticket

- Write a two-line program and what it will show.

Try it at school tomorrow
Play "Predict, then run": write three short programs on the board. Pairs write down what each one will show, then you run them on your phone and read the results aloud. Who was right, and why?

Practice questions · for you

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1 **A pupil types 12 when this program asks for a mark. What does input() give back?**

```
mark = input("Mark? ")
```

◯ Nothing until you print it ◯ The text "12" ◯ The number 12

2 **Each pupil needs 3 exercise books. Put the lines in order so the program asks how many pupils there are and prints how many books to buy. One line doesn't belong.**

Number the lines in order (1, 2, 3…). Cross out any line that doesn't belong.

☐ `pupils = int(text)`

☐ `text = input("How many pupils? ")`

☐ `books = text * 3`

☐ `books = pupils * 3`

☐ `print("Books to buy:", books)`

3 **After mark = 14 and then mark = 15, the variable mark holds both values.**

◯ True ◯ False

4 **Complete the program: it asks for a pupil's name and age, then says how old they will be next year.**

```
name = _____("Name? ")
age = _____(input("Age? "))
print(name, "will be", age + 1)
```

Word bank: `int` `str` `input` `print`

5 **Finish the program: it asks for two test marks and prints their average, like Average: 16.0.**

```
test1 = int(input("Test 1? "))
# ask for test 2, then print the average
```

Write your program:

✎ **Which job in your week could a few lines of Python do for you? Write the steps it would follow.**

Check your answers

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1 **The text "12"**

input() always gives back text, even when what was typed looks like a number.

2 **Top to bottom: 2, 1, X, 3, 4**

```
text = input("How many pupils? ")
pupils = int(text)
books = pupils * 3
print("Books to buy:", books)
```

Each line needs the one before it: the typed text, then the number made from it, then the sum, then the answer.

3 **False**

A variable holds one value at a time. The second line puts 15 into mark, and the 14 is gone.

4

```
name = input("Name? ")
age = int(input("Age? "))
print(name, "will be", age + 1)
```

input() asks the question and gives back text.

5

```
test1 = int(input("Test 1? "))
test2 = int(input("Test 2? "))
average = (test1 + test2) / 2
print("Average:", average)
```

Two inputs, each turned into a number, then the sum in brackets divided by 2.